

Submission DSA1-PS1-SUB06

Question DSA1-PS1-Q01

notation changes halfway:

1. $7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.
2. Comparison path: $4 \rightarrow 9 \rightarrow 6$; parent of 7 is 6.

marks, subscripts, and units are mixed up

result written as: path $4 \rightarrow 9 \rightarrow 6$; parent=6 (please interpret)

Question DSA1-PS1-Q02

notation changes halfway:

1. Maintain a hash set of values seen so far.
2. Before inserting x , if $x \in \text{seen}$ return True; otherwise insert x .
3. Expected $O(n)$ time and $O(n)$ auxiliary space; invariant: seen contains exactly the prior prefix.

marks, subscripts, and units are mixed up

result written as: hash set membership scan; expected $O(n)$ time, $O(n)$ space; prefix invariant (please interpret)

Question DSA1-PS1-Q03

notation changes halfway:

1. A balanced tree can be built recursively by choosing the median, with each key visited once: $\Theta(n)$.
2. Height is $\Theta(\log n)$, so search is $\Theta(\log n)$.

marks, subscripts, and units are mixed up

result written as: build $\Theta(n)$ with median recursion; search $\Theta(\log n)$ in balanced tree (please interpret)

Question DSA1-PS1-Q04

notation changes halfway:

1. Queue A; discover B,C; then dequeue B and discover D; dequeue C and discover E.
2. Order A,B,C,D,E; layer order equals shortest edge distance.

marks, subscripts, and units are mixed up

result written as: A,B,C,D,E; BFS explores by distance layers and yields fewest edges (please interpret)

Question DSA1-PS1-Q05

notation changes halfway:

1. Invariant: the prefix before index i is sorted and contains the original prefix values.
2. Initialization: one-element prefix is sorted; maintenance inserts the next item; termination covers the full array.

marks, subscripts, and units are mixed up

result written as: sorted prefix invariant; initialization, insertion maintenance, termination imply correctness (please interpret)

Question DSA1-PS1-Q06

notation changes halfway:

1. $7 > 4$, go right; $7 < 9$, go left; $7 > 6$, go right.
2. Comparison path: $4 \rightarrow 9 \rightarrow 6$; parent of 7 is 6.

marks, subscripts, and units are mixed up

result written as: path $4 \rightarrow 9 \rightarrow 6$; parent=6 (please interpret)